

"THE GUARDIAN": A MODULAR AI FOR PROACTIVE HUMAN-LAYER CYBERSECURITY ON SOCIAL MEDIA.

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Final Report

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Abstract

The rapid growth of social media platforms has introduced a new class of cybersecurity threats that primarily target human behavior rather than technical vulnerabilities. These human-layer threats—including misinformation, hate speech, social engineering scams, and the unintentional disclosure of personally identifiable information (PII)—exploit trust, cognitive biases, and contextual user behavior, posing significant risks to individuals and society. Unlike traditional cyberattacks, these threats are often embedded within natural language and rely on psychological manipulation and contextual inference, making them difficult to detect using conventional security mechanisms.

Existing protection approaches are largely reactive, platform-dependent, and designed to address isolated threat categories. Such solutions lack the ability to provide real-time, user-centric protection and are insufficient for identifying context-driven risks inherent in everyday social media interactions.

This research presents *The Guardian*, a modular AI-powered system designed to provide proactive, real-time assistance to users against human-layer cybersecurity threats on social media. The system is implemented as a browser extension integrated with a FastAPI-based backend and comprises four specialized Natural Language Processing (NLP) modules: misinformation detection, hate speech detection, scam and phishing detection, and PII and Operational Security (OpSec) risk detection. These modules leverage transformer-based models in combination with rule-based and heuristic techniques to analyze textual content and identify potential risks.

The system adopts a unified and context-aware approach, enabling the detection of both explicit threats and implicit risks arising from user behavior and content interaction. By integrating multiple detection capabilities within a single architecture, *The Guardian* provides real-time warnings, explanations, and guidance without restricting user actions. This user-centric design promotes informed decision-making while preserving usability.

Overall, the proposed system demonstrates the effectiveness of modular AI architectures in addressing diverse human-layer cybersecurity threats. The results highlight the importance of combining statistical learning with contextual reasoning to enhance real-time protection and user awareness in social media environments.

Keywords

Human-layer cybersecurity, social media security, natural language processing, misinformation detection, hate speech detection, phishing detection, scam detection, PII detection, operational security, explainable AI

1. Introduction

The rapid growth of social media platforms has significantly transformed how individuals communicate, share information, and interact in digital environments. However, this transformation has also led to the large-scale dissemination of harmful and misleading content, including misinformation, hate speech, scams, and privacy-threatening information leaks. These threats not only affect individuals but also contribute to broader societal and cybersecurity risks [1], [2].

Studies have shown that false information spreads faster and reaches wider audiences than truthful content, amplifying its impact on public opinion and decision-making processes [3]. At the same time, harmful online interactions such as hate speech and offensive language can negatively affect users' psychological well-being and contribute to unsafe digital environments [4]. Additionally, the increasing sophistication of online scams and phishing attacks demonstrates how adversaries exploit both technical vulnerabilities and human behavior through social engineering techniques [5].

From a cybersecurity perspective, these threats are increasingly recognized as **human-layer vulnerabilities**, where users themselves become targets due to cognitive biases, lack of awareness, and exposure to manipulated content [6]. Unlike traditional cyberattacks that exploit system-level weaknesses, human-layer threats operate through language, context, and behavioral cues, making them more difficult to detect and mitigate in real time.

Recent advancements in natural language processing (NLP), particularly transformer-based models such as BERT and its variants, have significantly improved the ability to analyze and interpret textual data in context [7], [8]. These models have been successfully applied to tasks such as misinformation detection, hate speech classification, and phishing detection. However, most existing approaches remain limited in scope, operating as standalone solutions that address individual threat categories without providing a unified or user-centric protection mechanism.

Furthermore, current systems often lack the ability to interpret context-driven risks and provide real-time feedback to users during interaction. As a result, users are not adequately supported in identifying complex or implicit threats embedded within everyday social media content.

To address these challenges, this research presents *The Guardian*, a modular AI-powered system designed to provide proactive, real-time assistance against multiple human-layer

cybersecurity threats on social media platforms. The system is implemented as a browser extension integrated with a FastAPI-based backend and incorporates four specialized Natural Language Processing (NLP) modules: misinformation detection, hate speech detection, scam and phishing detection, and personal information exposure detection.

The system adopts a unified architecture that integrates transformer-based models with rule-based and heuristic techniques, enabling the detection of both explicit threats and context-dependent risks. In addition, it emphasizes a user-centric approach by providing explanations and actionable guidance, allowing users to make informed decisions without restricting their interactions.

The key contributions of this work are as follows:

- A unified multi-module architecture for real-time detection of diverse human-layer cybersecurity threats
- Integration of machine learning, rule-based, and heuristic techniques within a single system framework
- A user-centric design that provides explainable feedback and actionable guidance
- A browser-based deployment enabling proactive protection during real-time user interaction

This research aims to bridge the gap between advanced AI-based detection techniques and practical, user-facing cybersecurity solutions, contributing to safer and more informed engagement in modern digital environments.

2. Related Work

The increasing prevalence of harmful content on social media has led to extensive research across multiple domains, including misinformation detection, hate speech classification, scam and phishing detection, and privacy risk analysis. These research areas highlight both technological challenges in content analysis and human-centered challenges related to user behavior, cognition, and decision-making in digital environments.

Early approaches to misinformation detection relied on traditional machine learning techniques using handcrafted features such as lexical patterns, syntactic structures, and metadata signals. While these methods achieved moderate success, they often struggled to generalize across datasets and failed to capture deeper contextual relationships within text [9]. The introduction of transformer-based models such as BERT significantly improved natural language understanding by enabling bidirectional contextual learning [7]. Lightweight variants such as DistilBERT further enhanced computational efficiency, making them more suitable for real-time applications [8]. Subsequent studies demonstrated that these models achieve high accuracy in fake news detection tasks across benchmark datasets [10]. However, despite their effectiveness, such approaches remain largely content-focused and often overlook the role of user perception and contextual interpretation.

In addition to content-based detection, research has emphasized the influence of human cognitive factors in misinformation spread. Studies show that false information propagates more rapidly than truthful content due to novelty and emotional engagement [3]. Furthermore, cognitive biases and limited analytical reasoning contribute to varying levels of susceptibility among users [11]. These findings suggest that effective solutions must go beyond detection accuracy and incorporate user-centered mechanisms that support informed decision-making.

Hate speech detection has also been widely explored within natural language processing. Early methods employed traditional classifiers such as Support Vector Machines and Logistic Regression, relying heavily on feature engineering [4]. While effective in controlled settings, these approaches were limited in capturing semantic nuance and contextual meaning. Deep learning architectures, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, improved performance by learning representations directly from data [12]. More recently, transformer-based models have achieved state-of-the-art performance by modeling long-range dependencies and contextual relationships [13].

Nevertheless, challenges such as dataset imbalance, annotation inconsistency, and context-dependent interpretation continue to limit their reliability in real-world applications.

Scam and phishing detection research has traditionally focused on URL-based and rule-based approaches, analyzing structural features such as domain characteristics, lexical patterns, and blacklist matching to identify malicious links [5]. Although effective for detecting known threats, these approaches often fail to identify newly generated or obfuscated attacks. To address this limitation, machine learning and deep learning techniques have been introduced, incorporating contextual analysis of textual content [14]. Transformer-based models have shown promising results in identifying social engineering patterns within messages [15]. However, such models can be computationally expensive and may introduce latency, making real-time deployment challenging.

To improve robustness, hybrid detection frameworks have been proposed, combining rule-based heuristics with machine learning models. These systems leverage the precision of deterministic rules and the contextual understanding of deep learning techniques [16]. Studies indicate that hybrid approaches outperform standalone methods, particularly in handling complex and context-dependent threats [17]. Despite these advancements, many hybrid systems are evaluated in offline settings and lack integration into user-facing environments where real-time interaction is required.

Another important area of research is privacy and personal information exposure detection. Existing approaches primarily focus on identifying structured PII, such as phone numbers and email addresses, using rule-based techniques, while unstructured entity detection is handled using Named Entity Recognition (NER) models [18]. However, these methods often fail to capture contextual risks, where seemingly benign information can be combined to infer sensitive details. This limitation highlights the need for approaches that consider not only explicit disclosures but also implicit, context-driven vulnerabilities.

Despite significant progress across these domains, several key research gaps remain. First, most existing solutions address individual threat categories in isolation, rather than providing a unified framework capable of handling multiple types of harmful content simultaneously. Second, many approaches operate in offline or platform-dependent environments, limiting their applicability in real-time user interactions. Third, existing systems often lack explainability and user-centric feedback mechanisms, reducing their effectiveness in influencing user behavior.

Finally, limited attention has been given to contextual and multi-signal risk inference, particularly in scenarios where threats emerge from the interaction of multiple weak signals.

To address these limitations, this research proposes a unified, real-time, and user-centric framework that integrates multiple detection modules within a browser-based environment. By combining transformer-based models with rule-based and heuristic techniques, and incorporating contextual analysis within a modular architecture, the proposed system aims to provide a comprehensive solution for detecting and mitigating diverse human-layer cybersecurity threats.

3. System Overview

The Guardian is a modular, AI-driven system designed to provide real-time protection against harmful and misleading content encountered on social media platforms. The system operates as a browser extension integrated with a backend processing framework, enabling seamless interaction between users and multiple detection modules.

At a high level, the system consists of two primary components: the client-side browser extension and the server-side backend. The browser extension is responsible for capturing user interactions, extracting relevant textual content, and presenting analysis results through a non-intrusive user interface. The backend processes the extracted data using a combination of machine learning models, rule-based techniques, and heuristic analysis to detect potential threats.

As illustrated in **Figure 1**, The Guardian follows a unified architecture in which user-generated content is routed through multiple detection modules before being aggregated into a single, interpretable output.

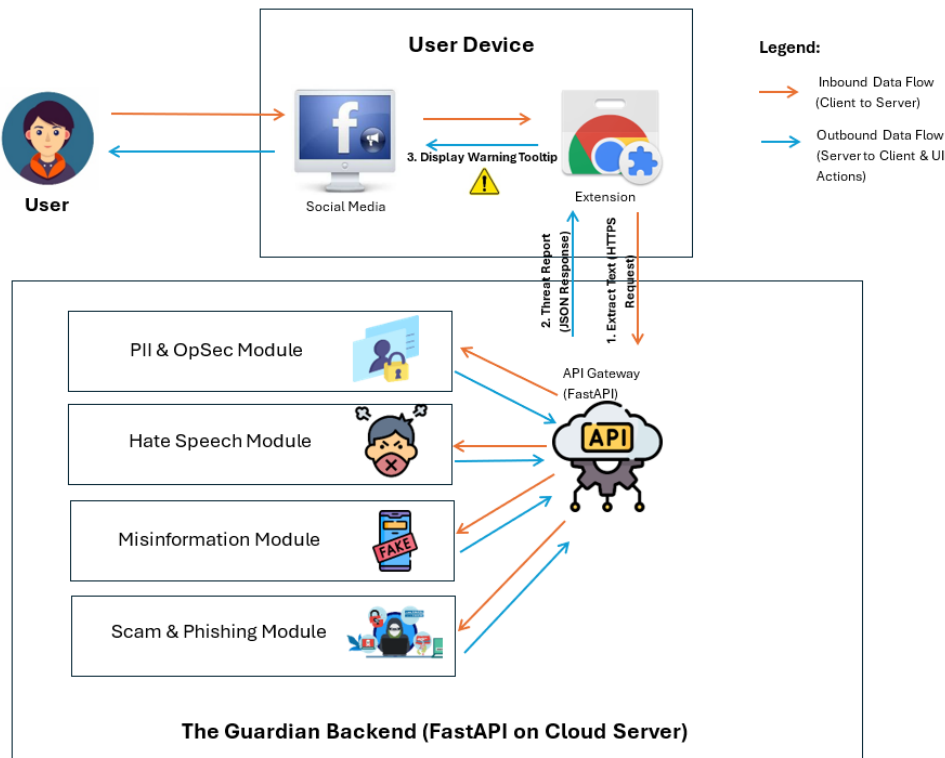


Figure 1: Overall Architecture of The Guardian System

The Guardian integrates four core detection modules, each targeting a specific category of harmful content:

- Misinformation Detection Module
- Hate Speech Detection Module
- Scam and Phishing Detection Module
- Personal Information Exposure Detection Module

These modules operate within a unified framework, enabling the system to analyze multiple dimensions of risk simultaneously. Unlike traditional approaches that treat these problems independently, the proposed system combines outputs from all modules to provide a holistic assessment of content safety.

Table 1: Summary of Detection Modules

Module	Purpose	Technique
Misinformation	Detect false/manipulative content	DistilBERT + heuristics
Hate Speech	Detect toxic language	DistilBERT + Transformer classifiers
Scam & Phishing	Detect social engineering attacks	URL + NLP hybrid
PII/OpSec	Detect privacy risks	Regex + NER + context

The system workflow begins when a user interacts with content on a social media platform. The browser extension extracts relevant text and sends it to the backend through a secure API request. The backend processes the input in parallel across all detection modules, each producing structured outputs such as classification labels, confidence scores, and contextual insights.

The outputs from all modules are then aggregated and transformed into a unified response, which is returned to the browser extension. The extension presents this information to the user through warnings, highlighted content, and explanatory feedback. This real-time interaction enables users to make informed decisions while engaging with online content.

A key design principle of The Guardian is its user-centric approach. Instead of blocking or censoring content, the system provides contextual explanations and actionable guidance, promoting awareness and responsible digital behavior. This design aligns with prior research emphasizing the importance of explainability and user involvement in AI-driven systems [19].

Overall, The Guardian bridges the gap between advanced AI-based detection techniques and practical cybersecurity solutions by delivering real-time, explainable, and multi-dimensional protection within a single integrated system.

4. System Architecture

4. System Architecture

The Guardian is implemented using a modular, API-driven architecture that enables real-time analysis of user-generated content across multiple threat domains. Building upon the system overview presented in Section 3, this section describes the internal processing pipeline, module integration strategy, and data flow mechanisms that support system functionality.

A. Client–Server Interaction

The system follows a client–server model, where the browser extension functions as the client and communicates with a backend service via HTTP requests.

The interaction process is summarized as follows:

1. The user interacts with content on a social media platform
2. The browser extension extracts relevant textual content or URLs
3. The extracted data is sent to the backend through a secure API request
4. The backend processes the input across multiple detection modules
5. A structured response is returned to the browser extension
6. The extension presents warnings, highlights, and explanatory feedback

This design enables real-time processing while maintaining a lightweight and non-intrusive client-side implementation.

B. Backend Processing Pipeline

The backend of The Guardian is implemented using the FastAPI framework, which serves as the central processing layer responsible for handling incoming requests and coordinating analysis across modules.

The processing pipeline consists of the following stages:

1) Input Preprocessing

Incoming text is normalized using standard natural language processing techniques, including:

- Tokenization
- Lowercasing
- Noise and symbol removal

These steps ensure consistency and compatibility with downstream analytical models.

2) Parallel Module Execution

The preprocessed input is passed simultaneously to all detection modules:

- Misinformation Detection
- Hate Speech Detection
- Scam and Phishing Detection
- Personal Information and OpSec Risk Detection

Each module operates independently and produces structured outputs such as classification labels, confidence scores, and contextual insights. Parallel execution improves efficiency and enables real-time response.

3) Feature Extraction and Analysis

Each module applies domain-specific analysis techniques tailored to its respective threat category. While misinformation, hate speech, and scam detection primarily rely on transformer-based classification models, additional techniques are employed to enhance detection capability.

These include:

- Transformer-based models (e.g., BERT, DistilBERT) for contextual understanding
- Rule-based and heuristic analysis for pattern-based detection
- Contextual feature extraction for capturing implicit signals

This combination allows the system to detect both explicit threats (e.g., offensive language, malicious URLs) and implicit risks (e.g., contextual cues or behavioral indicators embedded within text).

4) Result Aggregation

Outputs from all modules are aggregated into a unified representation. This process involves:

- Combining classification outputs from individual modules
- Resolving overlapping or conflicting detections

- Integrating entity-level and contextual signals
- Computing an overall risk level

The aggregation layer enables a coherent interpretation of multiple threat signals, allowing the system to provide a holistic assessment rather than isolated predictions.

5) Response Generation

The final output is formatted as a structured JSON response containing:

- Risk level (Low / Medium / High)
- Detected threat categories
- Confidence scores
- Highlighted text segments
- Explanatory feedback

This structured format ensures seamless integration with the browser extension and supports user-friendly presentation of results.

C. Module Integration Strategy

A key architectural feature of The Guardian is its loosely coupled module design, where each detection component operates independently while adhering to a standardized input-output interface.

This design provides several advantages:

- **Scalability** → New modules can be integrated without modifying existing components
- **Maintainability** → Individual modules can be updated or replaced independently
- **Parallel processing** → Reduces latency and improves responsiveness
- **Flexibility** → Supports integration of hybrid detection techniques

This modular architecture enables the system to evolve and adapt to emerging threats without requiring significant structural changes.

D. Real-Time Performance Considerations

To ensure practical usability, the system is optimized for low-latency processing:

- Lightweight transformer models (e.g., DistilBERT) are used where applicable
- Parallel module execution minimizes processing time
- Efficient API communication reduces overhead

These optimizations ensure that feedback is delivered quickly enough to support real-time user interaction.

E. Security and Privacy Considerations

The Guardian is designed with strong privacy and security principles. The system:

- Does not store user input permanently
- Uses secure communication protocols (HTTPS)
- Processes requests in a stateless manner

These measures ensure that user data remains protected while enabling effective analysis of content.

5. Methodology

The Guardian is designed as a modular, hybrid AI system for detecting and mitigating multiple forms of harmful social media content. The methodology integrates transformer-based natural language processing models with rule-based techniques and contextual reasoning mechanisms to enable accurate and real-time detection across diverse threat categories.

The design of the system is guided by three core principles:

- **Hybrid Intelligence** – Combining machine learning models with rule-based and heuristic techniques to improve robustness and reliability
- **Context Awareness** – Incorporating semantic and situational understanding beyond surface-level text to capture implicit risks
- **User-Centric Design** – Providing explainable outputs and actionable feedback to support informed user decision-making

Unlike traditional approaches that treat threats independently, The Guardian integrates multiple detection strategies within a unified framework, enabling comprehensive analysis of user interactions.

A. System-Level Methodology

At a system level, the methodology follows a multi-stage processing pipeline:

1. Data Acquisition

User-generated content is captured from social media platforms via the browser extension.

2. Preprocessing

The input is normalized using standard NLP techniques such as tokenization, text cleaning, and normalization.

3. Parallel Analysis

The processed input is analyzed simultaneously by multiple specialized detection modules.

4. Contextual Interpretation

Each module applies domain-specific techniques to identify patterns, risks, or anomalies based on both explicit and implicit signals.

5. **Aggregation and Scoring**

Outputs from all modules are combined to generate a unified risk assessment.

6. **Explanation Generation**

The system produces human-readable explanations and guidance to help users understand detected risks.

This pipeline enables the system to capture both explicit indicators (e.g., keywords, URLs) and implicit signals (e.g., tone, context, intent), supporting multi-dimensional threat detection.

B. PII and OpSec Risk Detection Module

The PII and Operational Security (OpSec) module is designed to detect both explicit and implicit privacy risks in user-generated content. While traditional approaches focus primarily on structured Personally Identifiable Information (PII), real-world privacy threats often arise from contextual disclosures that enable adversarial inference.

To address this challenge, the module adopts a hybrid, context-aware, and attacker-aware detection framework.

The detection process is structured into two complementary analytical paths:

• Path A — Explicit PII Detection

This path focuses on identifying structured and unstructured personal information using:

- Rule-based detection (regular expressions) for structured PII (e.g., phone numbers, email addresses, identification numbers)
- Named Entity Recognition (NER) models for unstructured entities such as names, locations, and organizations

This combination ensures high precision for well-defined patterns while maintaining generalization across natural language inputs.

• Path B — Contextual OpSec Signal Detection

This path captures implicit privacy risks that are not directly expressed as PII but can be inferred from contextual cues. These include:

- Future travel plans
- Absence from home

- Routine behaviors
- Vulnerability indicators

Contextual signals are extracted using a combination of rule-based heuristics and machine learning-based classification models.

A key observation from experimental analysis is that machine learning models alone exhibit limitations in handling ambiguity and boundary cases in contextual inference. This necessitates the integration of heuristic reasoning to improve robustness.

• **Attacker-Aware Risk Composition**

To address these limitations, the module incorporates an attacker-aware risk modeling layer. Rather than treating detected signals independently, this layer performs compositional reasoning by combining multiple signals to evaluate potential adversarial exploitation.

For example:

- Travel timing + absence indicators → potential physical targeting risk
- Routine patterns + location information → predictability vulnerability
- PII + contextual signals → increased identifiability and targeting risk

This compositional approach enables the detection of higher-order risks that are not captured by traditional PII detection systems.

Overall, this module extends privacy detection from simple entity recognition to contextual risk inference, aligning with real-world attacker behavior.

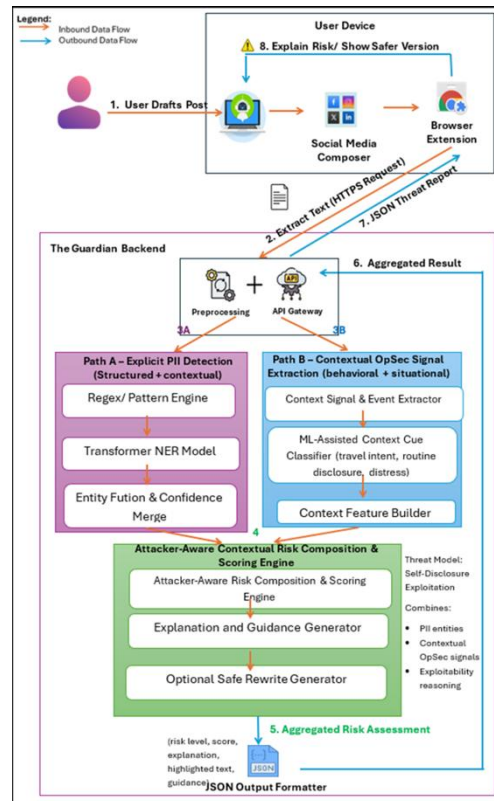


Figure 2: PII and OpSec Detection Architecture

C. Hate Speech Detection Module

The hate speech detection module utilizes transformer-based models to identify toxic and harmful language in social media content. Unlike traditional keyword-based approaches, this module emphasizes contextual and semantic understanding.

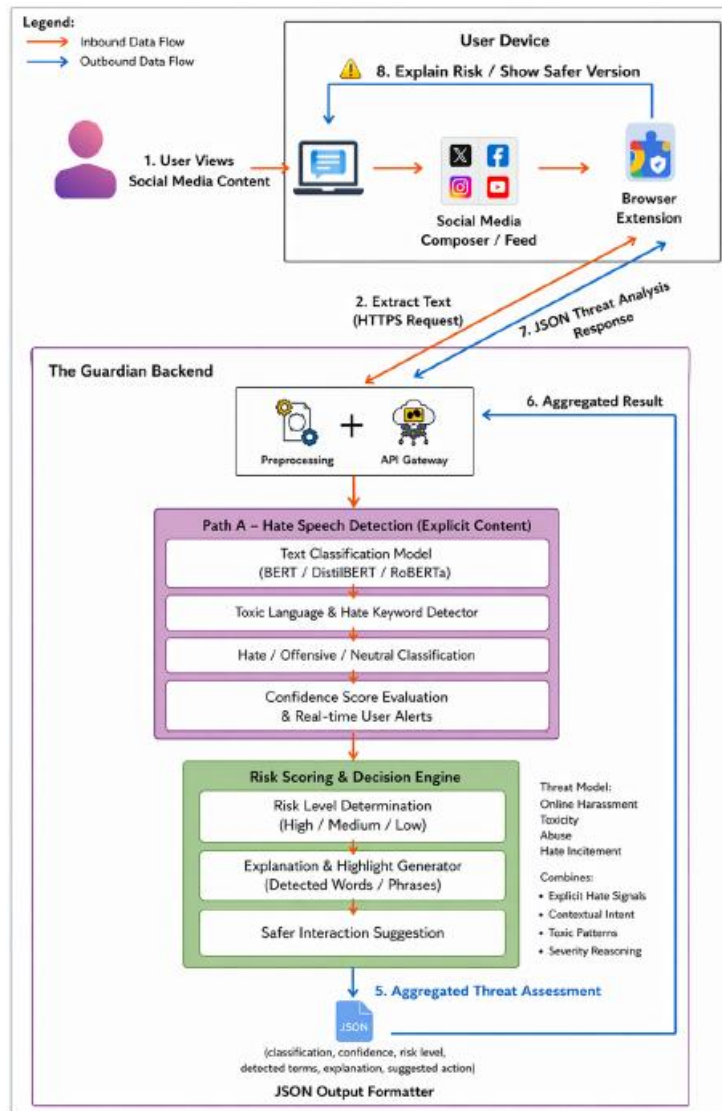
The detection pipeline consists of:

- Text preprocessing and tokenization
- Feature extraction using transformer-based embeddings
- Classification into categories such as hate speech, offensive language, or neutral content

To improve reliability, the module incorporates:

- Context-sensitive classification
- Reduction of false positives through semantic understanding
- Consideration of fairness and bias

This enables detection of both explicit and implicit forms of harmful language.



D. Misinformation Detection Module

The misinformation detection module focuses on identifying manipulative and misleading content, rather than strictly verifying factual correctness. It combines machine learning techniques with heuristic analysis to capture both structural and psychological aspects of misinformation.

The detection pipeline includes:

- Transformer-based classification using models such as DistilBERT
- Identification of manipulation techniques (e.g., emotional appeal, exaggeration)
- Contextual analysis of linguistic patterns

An important feature of this module is the incorporation of user-adaptive explanations, which provide tailored feedback to improve user understanding. This aligns with prior research showing that user perception significantly influences susceptibility to misinformation [20].

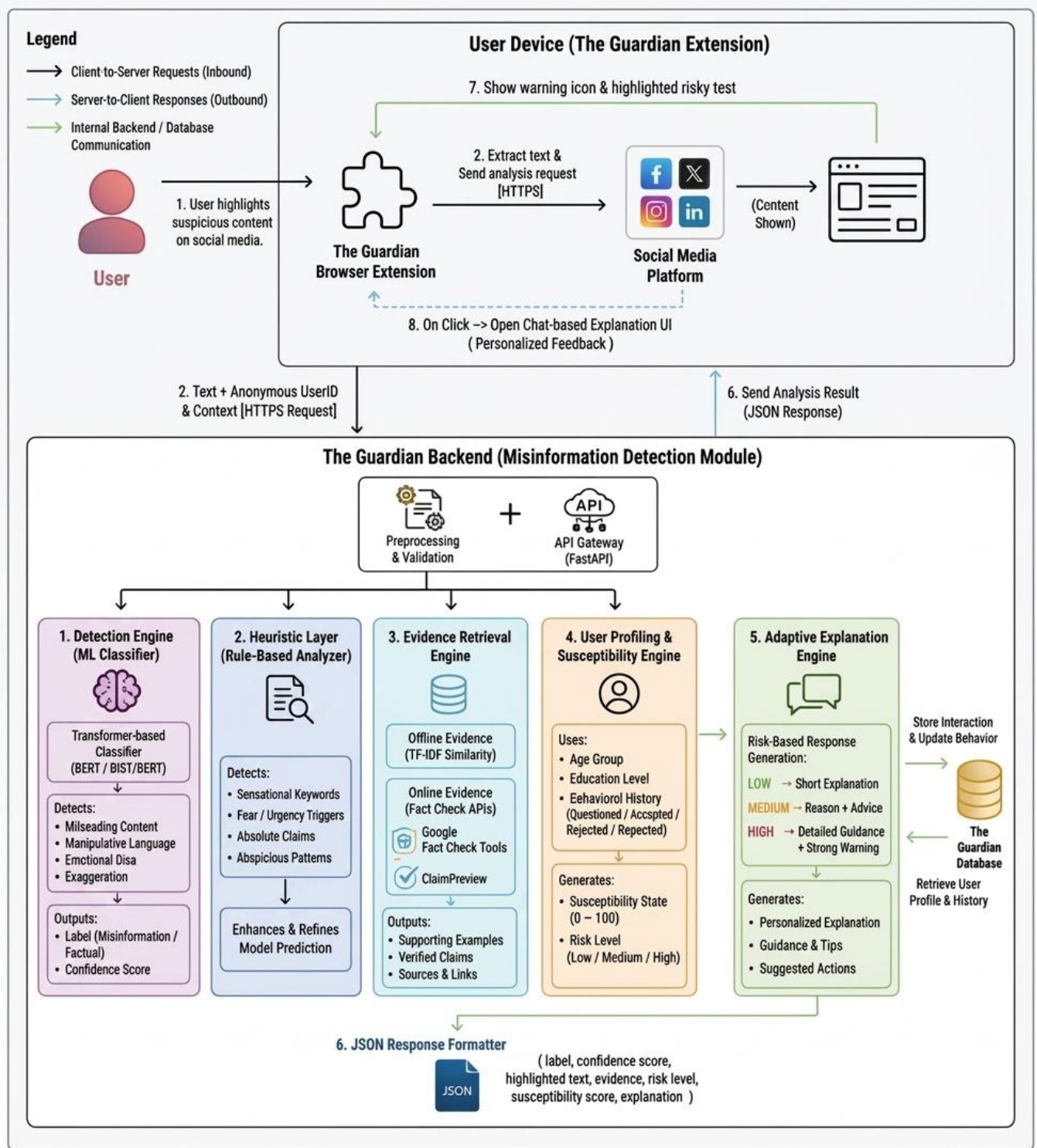


Figure 4: Misinformation Detection and Explanation Framework

E. Scam and Phishing Detection Module

The scam and phishing detection module is designed to identify social engineering attacks within social media environments, where malicious content is often embedded in informal, conversational text rather than structured formats such as emails or traditional phishing pages. Unlike conventional detection systems that rely solely on URL-based heuristics or static blacklists, this module adopts a hybrid, real-time approach that integrates both structural and contextual analysis to improve detection robustness.

The module operates using a dual-path detection strategy:

- **URL-based analysis**, which evaluates structural characteristics of links using heuristic rules and domain-based indicators
- **Text-based analysis**, which leverages transformer-based models to identify contextual scam patterns within user-generated content

This hybrid design enables the system to detect both explicit threats (e.g., malicious URLs) and implicit social engineering cues (e.g., urgency or impersonation), which are commonly used in modern scam campaigns.

1) Path A — Heuristic URL-Based Analysis

The first analytical path focuses on identifying structural indicators of malicious intent through rule-based URL analysis. This approach leverages domain-specific knowledge of phishing techniques and evaluates features such as:

- Suspicious lexical patterns (e.g., “login”, “verify”, “urgent”)
- URL length and structural complexity
- Use of IP addresses instead of domain names
- Presence of URL shortening services
- Abnormal or deceptive domain patterns

Each detected feature contributes to a cumulative risk score through a weighted heuristic framework. This deterministic approach provides high precision for known attack patterns and enables rapid detection with minimal computational overhead.

This path is particularly effective in detecting explicit threats embedded within URLs, especially in cases where textual context is limited or unavailable.

2) Path B — Contextual Text-Based Analysis

The second analytical path focuses on detecting implicit scam signals within textual content using transformer-based natural language processing models. A fine-tuned DistilBERT model is employed to capture semantic and contextual relationships within user-generated text.

This model identifies linguistic patterns commonly associated with social engineering attacks, including:

- Urgency and pressure tactics
- Impersonation of trusted entities
- Requests for sensitive or financial information
- Persuasive or emotionally manipulative language

Unlike traditional keyword-based approaches, the transformer model enables context-sensitive interpretation, allowing the system to detect subtle and indirect indicators of malicious intent.

3) Hybrid Risk Fusion Strategy

To combine the strengths of both analytical paths, the module employs a unified risk fusion mechanism. Let the input be defined as:

$$I = (T, U)$$

where T represents textual content and U represents the associated URL. Two independent risk scores are computed:

- Surl : risk score derived from heuristic URL analysis
- Stext : probability score derived from contextual text classification

The final risk score R is computed using a maximum-risk fusion strategy:

$$R = \max(S_{url}, S_{text})$$

This approach ensures that strong indicators from either structural or contextual analysis are preserved. In contrast to averaging-based methods, which may dilute high-risk signals, the maximum-risk strategy prioritizes security by retaining the most significant evidence of malicious intent. This design is particularly effective in adversarial scenarios where attackers attempt to obscure either structural or linguistic cues.

4) Context-Aware Scam Detection

A key aspect of this module is its ability to capture both explicit and implicit indicators of scams. While URL-based analysis detects direct threats, contextual analysis enables the identification of behavioral and psychological manipulation techniques used in social engineering.

By integrating these signals, the system can detect:

- Explicit threats (e.g., malicious or deceptive URLs)
- Implicit threats (e.g., persuasive scam narratives)
- Combined threats where multiple weak signals indicate higher risk

This multi-dimensional detection capability aligns with real-world attack patterns, where scams often rely on a combination of structural deception and contextual manipulation.

5) Methodological Significance

The proposed hybrid methodology addresses key limitations of existing scam detection approaches. Traditional URL-based systems lack contextual understanding, while purely machine learning-based approaches may fail to capture structural anomalies or operate efficiently in real-time environments.

By combining rule-based heuristics with transformer-based contextual analysis, the module achieves:

- Improved detection robustness across diverse attack types
- Enhanced ability to detect context-dependent social engineering attacks
- Greater resilience to obfuscation and evolving scam strategies

This hybrid design reflects the broader system objective of providing accurate, real-time, and user-centric protection against human-layer cybersecurity threats.

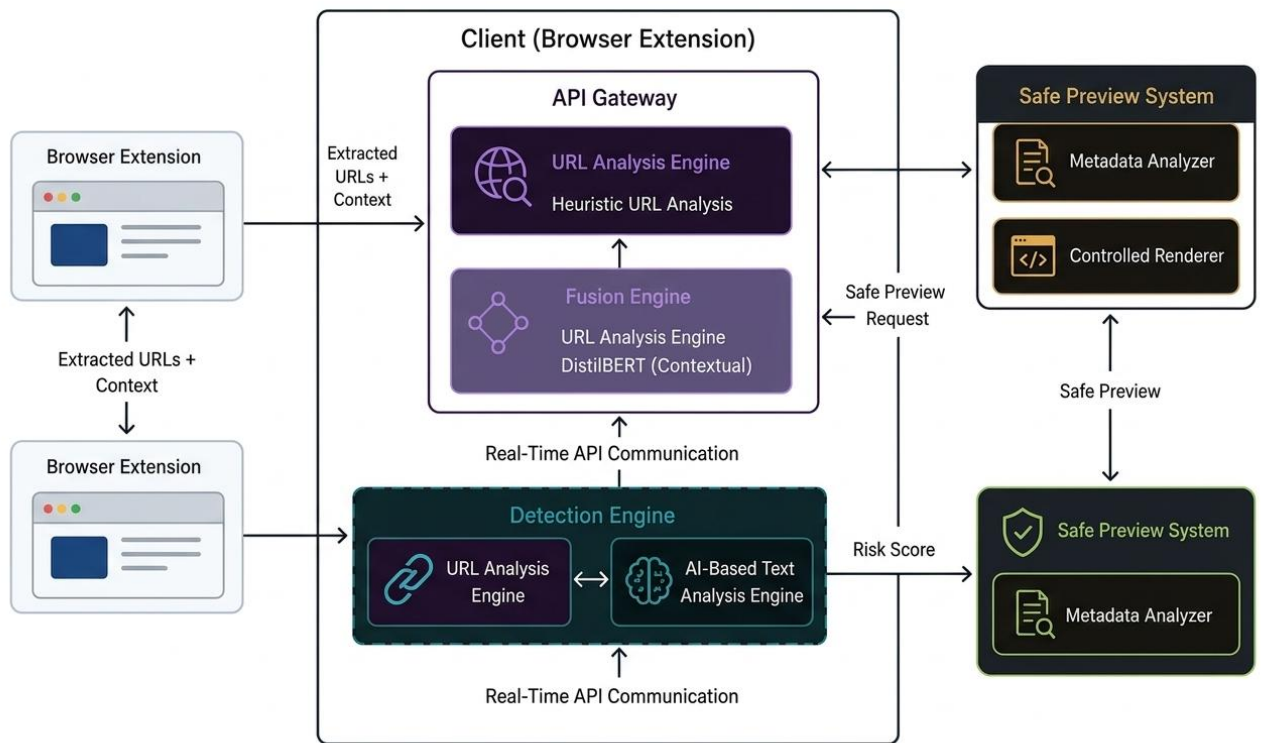


Figure 5: Scam and Phishing Detection Architecture

F. Integrated Risk Assessment

The outputs from all detection modules are combined to produce a unified risk assessment.

This stage involves:

- Aggregating detection results across modules
- Assigning an overall risk level
- Generating explanations and guidance

This integration enables the system to capture multi-dimensional threats, where multiple risk factors coexist within a single piece of content, providing a comprehensive and user-interpretable assessment.

6. Dataset Design and Preparation

The effectiveness of *The Guardian* system depends heavily on the quality, diversity, and realism of the datasets used for training and evaluation. Given the multi-module architecture, different dataset strategies were adopted for each detection task. While misinformation, hate speech, and phishing detection leveraged existing benchmark datasets, the PII and Operational Security (OpSec) module required extensive dataset construction and refinement due to the absence of suitable publicly available datasets for contextual privacy risk detection.

A. Dataset Overview

The dataset design process followed three key principles:

- **Task-specific dataset selection** for each module
- **Contextual realism**, particularly for human-layer threats
- **Balanced and leakage-free evaluation design**

Table 6.1 summarizes the datasets used across modules.

Table 6.1: Dataset Summary by Module

Module	Dataset Source	Type	Purpose
Misinformation	LIAR, FakeNewsNet	Public	Fake news classification
Hate Speech	Public social media datasets	Public	Toxic language detection
Scam & Phishing	URL datasets + message samples	Public + custom	Social engineering detection
PII Detection	PANORAMA	Synthetic	Structured & unstructured PII
OpSec Detection	Custom-built dataset	Synthetic + curated	Contextual risk inference

B. PII and OpSec Dataset

B.1 PANORAMA Dataset for PII Detection

For PII detection, the PANORAMA dataset was used as the primary data source. PANORAMA is a large-scale synthetic dataset designed to simulate realistic online user behavior and includes approximately 384,789 text samples generated from 9,674 synthetic user profiles across multiple content types such as social media posts, forums, and reviews.

The dataset is particularly suitable for PII detection as it contains diverse structured and unstructured personal information embedded in natural language contexts.

B.2 PII Dataset Construction

Since PANORAMA is not directly annotated for token-level PII recognition, a hybrid labeling strategy was applied:

- **Rule-based labeling (Regex):**
 - Phone numbers
 - Email addresses
 - National identity numbers
- **NER-based labeling:**
 - Person names
 - Locations
 - Organizations

This hybrid labeling approach aligns with prior research indicating that structured identifiers are best captured through deterministic rules, while contextual entities require machine learning-based recognition.

The dataset was then split using a **70/15/15 train-validation-test strategy**.

Figure 6.2: Sample with Entities after Split

Split	Samples with PII
Train	54,754
Validation	11,700

Test	11,684
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B.3 OpSec Problem Formulation and Threat Model

Unlike PII detection, OpSec risk detection is not an entity recognition problem. Instead, it is formulated as a **multi-label contextual inference task**, where risk emerges from behavioral and contextual signals rather than explicit identifiers.

This work adopts an **attacker-aware threat model**, where risk is defined based on how information can be exploited rather than its surface sensitivity. This aligns with human-layer cybersecurity principles, where attackers leverage user behavior and contextual disclosures [6].

The system considers multiple attacker types:

- Opportunistic attackers
- Targeted attackers / stalkers
- Social engineers
- Physical intruders

Based on this framework, the following OpSec label schema was defined:

Table 6.3: OpSec Label Schema

Label	Description
OPSEC_TRAVEL_FUTURE	Future travel implying absence
OPSEC_ABSENCE_FROM_HOME	Home may be unattended
OPSEC_ROUTINE_DISCLOSURE	Predictable behavior patterns
OPSEC_VULNERABILITY_SIGNAL	Emotional or situational vulnerability
OPSEC_LOCATION_EXPOSURE	General location clues
OPSEC_REALTIME_LOCATION	Current presence at a location

B.4 Rule-Based Baseline Design

Before applying machine learning models, a refined rule-based baseline system was implemented. This baseline introduced:

- **Label-level aggregation**
- **Priority-based trigger resolution**
- **Context-aware gating**
- **Multi-label detection**

This step was critical for establishing a **strong interpretable baseline** and enabling meaningful comparison with machine learning models.

B.5 OpSec Dataset Construction

Due to the lack of suitable datasets, a synthetic dataset was constructed using controlled generation strategies. The dataset includes:

- Balanced representation of all OpSec categories
- Multi-label scenarios
- Safe (non-risk) examples

The final dataset consists of **2,888 samples**, ensuring both diversity and balance.

Table 6.4: OpSec Dataset Distribution

Category	Count
Travel Future	534
Absence from Home	502
Routine Disclosure	553
Vulnerability Signal	490
Real-Time Location	498
Location Exposure	448
Multi-label	482
Safe Negative	345

B.6 Dataset Hardening

Initial experiments revealed that models were learning **template patterns rather than contextual meaning**, leading to inflated performance.

To address this, a **semantic hardening process** was applied:

- Introduction of ambiguous and neutral cases
- Weak vs strong signal separation
- Template rewriting
- Confusable boundary cases

This reduced overfitting and improved dataset realism.

B.7 Leakage-Free Dataset Splitting

A major issue in early experiments was **dataset leakage**, where similar samples appeared across training and validation sets.

To address this, a **group-aware splitting strategy** was implemented:

- Train: 70%
- Validation: 15%
- Test: 15%

This ensured that semantically similar samples did not appear across splits, improving evaluation reliability.

B.8 Challenge Dataset and Robustness Testing

To evaluate real-world performance, a separate **challenge dataset** was constructed containing:

- Ambiguous cases
- Indirect disclosures
- Boundary-confusable examples
- Multi-label hard cases

This dataset revealed that:

- Models perform well on strong signals

- Performance drops under ambiguity
- Failures are **task-driven rather than model-specific**

B.9 Summary of Key Findings

The dataset design process revealed several important insights:

- Contextual privacy risk is fundamentally an **inference problem**, not a detection problem
- Dataset quality and realism significantly impact model performance
- Ambiguity is the primary challenge in OpSec detection
- Hybrid approaches are necessary to handle real-world variability

C. Misinformation Dataset

The misinformation detection module was trained and evaluated using publicly available benchmark datasets, including LIAR and FakeNewsNet. These datasets contain labeled claims and news articles annotated for truthfulness and credibility. They provide a balanced mix of factual and misleading content, enabling the model to learn patterns associated with manipulation, exaggeration, and misinformation. Preprocessing steps included text normalization, tokenization, and removal of irrelevant metadata to ensure consistency with transformer-based input requirements.

D. Hate Speech Dataset

The hate speech detection module utilizes labeled social media datasets containing annotations for hate speech, offensive language, and neutral content. These datasets capture a wide range of linguistic variations, including explicit hate expressions and implicit contextual toxicity. Preprocessing included tokenization and normalization, while care was taken to address dataset imbalance and labeling inconsistencies commonly observed in hate speech corpora.

E. Scam and Phishing Dataset

The scam and phishing detection module requires a dataset that captures both **structural URL-based threats** and **contextual social engineering patterns**. To achieve this, a hybrid dataset

was constructed by combining multiple sources of scam-related textual data and phishing URL datasets.

The final dataset consists of **28,436 samples**, including both scam and non-scam instances. The data reflects real-world scenarios such as phishing messages, fraudulent promotions, impersonation attempts, and benign user-generated content. The dataset is inherently imbalanced, with approximately:

- **87.8% non-scam samples**
- **12.2% scam samples**

This imbalance reflects real-world distributions, where malicious content is relatively rare compared to benign content.

1) Data Preprocessing

To ensure compatibility with transformer-based models, the dataset undergoes several preprocessing steps:

- Text normalization (lowercasing, noise removal)
- Tokenization for model input
- Replacement of URLs with a placeholder token

The URL replacement strategy prevents the model from memorizing specific malicious links and encourages learning of generalized contextual patterns associated with scam behavior.

2) Handling Class Imbalance

Given the skewed class distribution, a **weighted cross-entropy loss function** is applied during model training. This assigns higher importance to the minority class (scam instances), improving recall and ensuring that malicious content is not overlooked.

This approach is particularly important in cybersecurity applications, where false negatives (missed scams) pose a greater risk than false positives.

3) URL Dataset Integration

In addition to textual data, the module incorporates URL-based datasets to support heuristic analysis. These include:

- A whitelist of trusted domains
- A blacklist of known malicious URLs

These datasets enable the rule-based component to detect structural anomalies and known attack patterns, complementing the contextual capabilities of the machine learning model.

7. Implementation

The Guardian system was implemented as a browser-based client application integrated with a backend API service, enabling real-time analysis of user-generated content on social media platforms. The implementation emphasizes modularity, scalability, and low-latency processing to support simultaneous detection of multiple threat categories.

A. System Environment and Technology Stack

The system was developed using a combination of modern web technologies and machine learning frameworks. The core components include:

- **Frontend (Browser Extension):**

Implemented using JavaScript, HTML, and CSS, the extension operates within the browser environment to capture user interactions and present analysis results.

- **Backend (API Service):**

Developed using the FastAPI framework in Python, providing a lightweight and high-performance interface for processing requests and coordinating module execution.

- **Machine Learning Frameworks:**

Transformer-based models were implemented using libraries such as Hugging Face Transformers and PyTorch, enabling efficient loading, fine-tuning, and inference of pre-trained models.

- **Data Processing Libraries:**

Python libraries such as NumPy and Pandas were used for data preprocessing, transformation, and analysis.

This technology stack enables efficient communication between components while supporting real-time inference and modular integration.

B. Browser Extension Implementation

The browser extension serves as the user-facing component of The Guardian. Its primary responsibilities include:

- Capturing user-generated or highlighted content from social media platforms
- Sending extracted data to the backend API
- Displaying detection results through visual indicators and explanations

The extension consists of the following components:

- **Content Script:** Extracts relevant textual content from web pages
- **Background Script:** Manages communication with the backend API
- **Popup Interface:** Displays warnings, explanations, and interaction options

The extension is designed to be lightweight and non-intrusive, ensuring minimal disruption to the user experience while enabling real-time feedback.

C. Backend API Implementation

The backend API acts as the central processing unit of the system and is implemented using FastAPI due to its asynchronous capabilities and high performance.

The primary endpoint used in the system is:

POST /analyze

This endpoint receives user input and returns structured analysis results.

The backend is organized into modular components:

- **Request Handler:** Validates and processes incoming data
- **Processing Pipeline:** Routes input to detection modules
- **Aggregation Layer:** Combines outputs from all modules
- **Response Generator:** Formats results into structured JSON responses

This modular backend design enables efficient handling of concurrent requests, which is essential for real-time operation.

D. Module Integration

Each detection module is implemented as an independent processing unit within the backend.

This modular design allows components to be developed, evaluated, and updated independently.

Key integration features include:

- **Parallel execution of modules** to reduce latency
- **Standardized input/output interfaces** for interoperability
- **Flexible routing mechanisms** for handling different input types

Additionally, the integration layer supports the combination of outputs from heterogeneous models and techniques, including transformer-based classifiers, rule-based detection, and contextual signal analysis. This enables the system to leverage both statistical learning and deterministic reasoning within a unified framework.

E. Model Implementation

The system utilizes pre-trained transformer models that are fine-tuned for specific tasks such as classification and entity recognition.

The implementation process includes:

1. Loading pre-trained models (e.g., BERT, DistilBERT)
2. Fine-tuning models on domain-specific datasets
3. Evaluating performance using validation datasets
4. Deploying trained models for inference within the backend pipeline

In addition to transformer-based models, rule-based and heuristic components are integrated where necessary to improve detection performance, particularly for structured patterns and context-dependent signals.

To optimize performance, lightweight models such as DistilBERT are used where appropriate, balancing accuracy and computational efficiency [8].

F. Data Flow Implementation

The system follows a structured data flow from input to output:

1. User input is captured by the browser extension
2. The input is transmitted to the backend API
3. The backend preprocesses and normalizes the data
4. Detection modules analyze the input in parallel
5. Outputs are aggregated into a unified representation
6. A structured response is returned to the extension
7. The extension displays results to the user

This pipeline ensures efficient and consistent processing across all modules while supporting real-time interaction.

G. Performance Considerations

Several optimizations were implemented to ensure real-time performance:

- Use of lightweight transformer models
- Parallel execution of detection modules
- Efficient API communication
- Minimal preprocessing overhead

These optimizations reduce latency and ensure that users receive timely feedback during interaction.

H. Deployment Considerations

The system was designed to support flexible deployment configurations, including:

- Local environments for development and testing
- Cloud-based deployment for scalability and availability
- Browser extension distribution for end-user accessibility

The modular architecture allows individual components to be deployed, scaled, or updated independently, supporting long-term maintainability and extensibility.

E. Scam and Phishing Module Implementation

The scam and phishing detection module is implemented using a hybrid architecture that combines heuristic URL analysis with transformer-based contextual classification.

1) Heuristic URL Engine

The URL analysis engine is implemented as a rule-based system that evaluates structural features of URLs. It uses predefined rules to detect patterns such as suspicious keywords, abnormal domain structures, and the use of URL shorteners.

Each detected feature contributes to a cumulative risk score, enabling fast and interpretable detection of known phishing patterns.

2) Text Classification Model

The contextual analysis component is implemented using a fine-tuned DistilBERT model for binary classification (scam vs non-scam). The model is trained on a domain-specific dataset and optimized for real-time inference.

Key implementation details include:

- Pre-trained model: **distilbert-base-uncased**
- Framework: Hugging Face Transformers (PyTorch)
- Training configuration:
 - Learning rate: 5e-5
 - Batch size: 16
 - Epochs: 3
 - Maximum sequence length: 128

To address class imbalance, a weighted cross-entropy loss function is applied, improving recall for scam detection.

3) Hybrid Fusion Implementation

The outputs of the heuristic engine and the transformer model are combined using a maximum-risk fusion strategy. This ensures that high-risk signals from either component are retained in the final decision.

The fusion logic is implemented within the aggregation layer, producing a unified risk score used for classification.

4) Safe Preview System

The Safe Preview feature is implemented as a controlled interaction mechanism that allows users to inspect suspicious links without direct exposure.

The system includes:

- **Metadata Analyzer:** Extracts domain-level information

- **Static Renderer:** Displays content without executing scripts
- **Sandbox Environment (optional):** Isolates full rendering

This feature enhances user safety by preventing execution of potentially malicious content.

F. Data Flow Implementation

The system follows a structured data flow pipeline:

1. User interacts with content on a webpage
2. Browser extension extracts text and URLs
3. Data is sent to the backend API
4. Input is preprocessed and normalized
5. Detection modules analyze input in parallel
6. Outputs are aggregated into a unified risk assessment
7. Results are returned to the extension
8. User receives real-time feedback

This pipeline ensures consistent and efficient processing across all modules.

8. Evaluation and Results

The evaluation of *The Guardian* system focuses on assessing both the effectiveness of individual detection modules and the performance of the integrated system in realistic social media scenarios. Beyond standard classification accuracy, the evaluation emphasizes robustness, contextual understanding, and real-time usability, reflecting the challenges of human-layer cybersecurity.

A. Evaluation Objectives

The evaluation is designed to assess:

- Accuracy and reliability of individual detection modules
- Robustness under ambiguous and real-world conditions
- Effectiveness of the integrated multi-module architecture
- System performance in terms of latency and usability

These objectives align with the goal of developing a **real-time, user-centric cybersecurity system** capable of supporting decision-making during live user interactions.

B. Evaluation Datasets

The evaluation utilizes both publicly available datasets and custom-constructed datasets, as described in Section 6.

- **Misinformation:** LIAR, FakeNewsNet
- **Hate Speech:** Public labeled social media datasets
- **Scam & Phishing:** URL datasets and scam message corpora
- **PII & OpSec:** PANORAMA-based PII dataset and custom OpSec dataset with challenge sets

Data preprocessing included normalization, tokenization, and label standardization to ensure compatibility with transformer-based models.

C. Evaluation Metrics

The system was evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide a balanced evaluation of classification performance and are widely used in NLP tasks [25].

D. Module-Level Evaluation

1) Misinformation Detection

The misinformation module demonstrates high performance by combining transformer-based classification with heuristic analysis, enabling the detection of both factual inaccuracies and manipulative language patterns.

Table 2: Misinformation Detection Results

Metric	Value
Accuracy	98.85%
Precision	0.991
Recall	0.985
F1-score	0.988

2) Hate Speech Detection

The hate speech module demonstrated reliable performance in detecting toxic and offensive content using transformer-based classification.

Table 3: Hate Speech Detection Results

Metric	Value
Accuracy	92.01%

Precision	0.7834
Recall	0.7845
F1-score	0.7831

3) Scam and Phishing Detection

The scam and phishing module effectively identified social engineering patterns by combining URL-based analysis with NLP techniques.

Table 4: Scam and Phishing Detection Results

Metric	Value
Accuracy	98.63%
Precision (Weighted)	0.9874
Recall (Weighted)	0.9863
F1-score (Weighted)	0.9865

In addition, macro-averaged metrics are:

- Precision: 95.16%
- Recall: 98.84%
- F1-score: 96.90%

These results indicate strong performance across both majority and minority classes.

b) Class-Wise Performance

A detailed class-wise analysis highlights the system's effectiveness:

Safe Class (Non-Scam):

- Precision: 99.88%
- Recall: 98.56%
- F1-score: 99.21%

Scam Class (Critical Class):

- Precision: 90.45%
- Recall: 99.13%
- F1-score: 94.59%

The high recall for the scam class indicates that the system successfully detects the vast majority of malicious instances, which is critical for minimizing user risk.

c) Confusion Matrix Analysis

The confusion matrix results are as follows:

- True Positives (TP): 682
- True Negatives (TN): 4923
- False Positives (FP): 72
- False Negatives (FN): 6

The low number of false negatives demonstrates the system’s effectiveness in identifying malicious content, which is essential in cybersecurity applications.

d) Hybrid Model Effectiveness

To evaluate the contribution of the hybrid approach, the model is compared with baseline methods:

Model	Accuracy	Precision	Recall (Scam)	F1-score
URL-Based Heuristic	91.2%	89.5%	85.3%	87.3%
Traditional ML	94.8%	93.2%	91.0%	92.1%
Deep Learning (BERT)	97.2%	96.5%	95.8%	96.1%
Guardian (Hybrid)	98.63%	98.74%	99.13%	98.65%

The results show that the hybrid model outperforms standalone approaches, particularly in recall, demonstrating its effectiveness in detecting both structural and contextual scam indicators.

e) Real-Time Performance

The system achieves real-time performance with an average latency of **150–250 ms per request**.

Component	Time
URL Analysis	< 10 ms
AI Inference	100–200 ms
Total	150–250 ms

This ensures immediate feedback without disrupting user interaction.

f) Security-Oriented Analysis

From a cybersecurity perspective, recall is the most critical metric for scam detection. The system achieves a scam recall of **99.13%**, indicating that almost all malicious instances are detected.

Although false positives are present, this trade-off is acceptable in security-sensitive systems, where missing a threat is more harmful than incorrectly flagging benign content.

4) PII and OpSec Detection

The PII and OpSec module is evaluated using a **multi-level evaluation framework**, reflecting the complexity of both structured entity detection and contextual risk inference.

a) Model Comparison (PII Detection)

Table 5: PII Detection Model Performance Comparison

Model	Precision	Recall	F1-score	Latency (ms)	Key Insight

spaCy (Full)	0.864	0.868	0.866	18.39	Best for structured PII
spaCy (Subset)	0.849	0.825	0.837	18.27	Performance drop with less data
DistilBERT	0.666	0.600	0.632	52.55	Weak for structured PII
RoBERTa	0.667	0.624	0.645	99.53	Better context, weak phone
BERT	0.664	0.620	0.641	98.40	Balanced but weaker

The results indicate that **rule-based and NER approaches outperform transformer models for structured PII**, due to the deterministic nature of patterns such as phone numbers and emails.

b) Entity-Level Performance

Table 6: Per-Entity Performance of spaCy Model

Entity	F1-score	Interpretation
PHONE	0.9512	Excellent
EMAIL	0.9966	Near perfect
PERSON	0.8744	Strong contextual detection
LOCATION	0.8765	Strong
ORG	0.8224	Good
NIC	0.000	Not learned due to data limitation

This highlights the importance of **dataset coverage**, as underrepresented entities significantly affect model performance.

c) Challenge-Based Evaluation (OpSec)

Table 7: Challenge Evaluation Performance

Model	Exact Match	Micro F1	Macro F1	Observation
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DistilBERT	0.8077	0.8387	0.8421	Stable under ambiguity
BERT	0.7885	0.8308	0.8399	Slightly less robust
RoBERTa	0.7500	0.8060	0.8087	Most sensitive

Unlike PII detection, OpSec evaluation reveals that **performance degradation occurs across all models under ambiguity**. This confirms that the limitation lies in the **inherent semantic complexity of contextual inference**, rather than model architecture.

This finding aligns with prior research on contextual privacy and inference risks, which emphasizes that meaning extraction depends heavily on situational context rather than explicit signals [22], [23].

d) System-Level PII + OpSec Evaluation

Table 8: Category-wise System Performance

Category	Accuracy (%)
Safe	40%
PII	80%
OpSec	80%
Combined	100%
Ambiguous	80%

The system performs best in **multi-signal scenarios**, demonstrating the effectiveness of signal aggregation and compositional reasoning.

e) Error Analysis

Table 9: Error Type Summary

Error Type	Description
Over-trigger	Safe content incorrectly flagged

Ambiguity misinterpretation	Context confusion
Weak signal detection	Partial inference failure
Boundary confusion	Difficult edge cases

Error analysis shows that failures are primarily driven by **ambiguity and weak signals**, reinforcing the need for hybrid and reasoning-based approaches.

E. System-Level Qualitative Evaluation

Due to the multi-domain nature of *The Guardian* system, conducting a unified quantitative evaluation across all modules is not straightforward. Each module operates on distinct datasets with different labeling schemes and objectives, making the construction of a single ground-truth dataset impractical.

Instead, a **qualitative scenario-based evaluation** was conducted to assess the effectiveness of the integrated system in realistic user interactions.

The evaluation focused on:

- Detection of multiple threat types within a single input
- Consistency of aggregated risk assessment
- Clarity and usefulness of explanations
- User-facing interpretability and guidance

Scenario-Based Analysis

Several real-world inspired scenarios were tested:

1. Multi-risk Scenario

Example:

“I’m flying to Japan tomorrow, no one is at home, call me at 07XXXXXXXX”

Detected Risks:

- PII: Phone number
- OpSec: Travel + absence

- Combined risk: HIGH

2. Manipulative Content Scenario

Example:

“This miracle cure works instantly — doctors hate it!”

Detected Risks:

- Misinformation indicators
- Emotional manipulation patterns

3. Social Engineering Scenario

Example:

“Your bank account is locked. Click here immediately.”

Detected Risks:

- Scam/phishing patterns
- Urgency + impersonation cues

4. Toxic Language Scenario

Example:

“People like you don’t belong here.”

Detected Risks:

- Hate speech classification

Key Observations

The qualitative evaluation demonstrates that:

- The system successfully identifies **multiple concurrent threats** within a single input
- Risk aggregation enables **multi-dimensional understanding** of content
- Explanations improve interpretability and support user decision-making
- The modular architecture allows seamless integration of heterogeneous detection mechanisms

Limitations of System-Level Evaluation

- Lack of unified labeled datasets across multiple threat categories
- Difficulty in defining a single ground-truth label for multi-risk scenarios
- Evaluation relies on scenario-based testing rather than large-scale benchmarks

9. Discussion and Limitations

The Guardian demonstrates the feasibility of a modular, AI-assisted browser-based system for detecting multiple human-layer cybersecurity threats in social media environments. The results show that transformer-based models are effective for several NLP classification tasks, while hybrid approaches combining machine learning with rule-based and heuristic techniques are particularly useful for risks involving structured patterns, contextual cues, and attacker-relevant inference.

A major strength of the system is its modular design. Since misinformation, hate speech, scam/phishing, and PII/OpSec detection are implemented as separate components, each module can be developed, tested, improved, and replaced independently. This supports scalability and future extension, which was one of the original design goals of The Guardian. The project proposal also identified modularity, real-time operation, user guidance, and platform independence as core differentiators of the system when compared with reactive platform moderation and single-threat tools.

Another important contribution is the user-centric design. Unlike moderation systems that block or remove content, The Guardian provides warnings, explanations, and guidance while preserving user autonomy. This approach is suitable for human-layer cybersecurity because the goal is not only detection, but also user awareness and safer decision-making. The original proposal framed the browser extension as a real-time guide rather than a censorship mechanism, where the extension extracts social media text, sends it to the backend modules, and returns a subtle warning icon with tooltip guidance.

The PII and OpSec module further highlights the value of hybrid reasoning. Structured PII such as phone numbers and email addresses benefits from deterministic pattern detection, while unstructured entities and contextual privacy risks require NER, transformer-based classification, and attacker-aware reasoning. The project requirements for the PII/OpSec module explicitly included real-time scanning, structured PII detection, unstructured PII detection, contextual OpSec risk analysis, JSON output generation, and non-intrusive UI alerts. These requirements justify the hybrid design used in the final module.

However, several limitations remain.

A. Dataset Limitations

The performance of each module depends heavily on the quality and representativeness of its dataset. Public benchmark datasets such as LIAR, FakeNewsNet, Davidson, HateXplain, phishing URL datasets, and SMS scam datasets provide a useful starting point, but they do not perfectly reflect live social media environments. Social media posts are often short, informal, multilingual, sarcastic, emoji-rich, and context-dependent. As a result, models trained on benchmark datasets may not generalize equally well to all real-world posts.

For the PII/OpSec module, PANORAMA provided a strong synthetic foundation for PII-rich online text, but synthetic datasets may still contain generation bias, repeated patterns, or reduced linguistic diversity. Similarly, the custom OpSec dataset was necessary because no suitable public dataset existed for attacker-aware contextual risk inference. However, since the OpSec dataset was synthetically and manually constructed, its coverage is limited by the scenarios anticipated during dataset design.

B. Limited Real-World User Evaluation

The current evaluation includes module-level quantitative testing and qualitative scenario-based analysis. However, a full-scale user study was not conducted across a diverse population. This limits the ability to measure how real users interpret warnings, whether alerts change behavior, and whether repeated warnings lead to alert fatigue.

A stronger future evaluation should include controlled user testing, surveys, and usability metrics such as clarity, perceived usefulness, trust, response time acceptability, and willingness to follow safe rewrite suggestions. The initial project plan included qualitative user feedback collection as part of testing and evaluation, but the final system-level user study remains a future improvement rather than a completed component.

C. Scope and Platform Limitations

The Guardian was designed primarily for social media environments, with early focus on platforms such as Facebook and X/Twitter. This scope is reasonable for the project because social media is a major source of human-layer threats. However, it also limits

generalization to other environments such as WhatsApp, Telegram, email, Discord, LinkedIn, Reddit, and comment sections on news platforms.

The original system flow focused on a browser extension extracting visible text from Facebook or Twitter and sending it to the FastAPI backend for analysis. Therefore, the current implementation should be understood as a browser-based social media prototype, not a universal cross-platform security product.

D. Edge Case Limitations

The system may struggle with edge cases such as sarcasm, coded language, memes, mixed-language text, deliberate obfuscation, and evolving slang. Hate speech detection may misclassify satire, quoted speech, reclaimed language, or coded “dog whistle” phrases. Misinformation detection may struggle when claims require external verification or domain-specific expertise. Scam detection may fail when attackers use novel persuasion strategies or links hidden behind redirects and shortened URLs. PII/OpSec detection may over-trigger on ambiguous phrases or miss weak contextual signals.

These limitations are especially visible in OpSec detection, where the core problem is not only classification but contextual inference. Ambiguous statements such as “still here,” “might head out,” or “this feels uncomfortable” can be difficult to classify because their risk depends on implied context, user history, location, and attacker perspective.

E. Risk of False Positives and Over-Triggering

The system intentionally prioritizes user safety, but this can lead to false positives. In security-sensitive systems, high sensitivity can reduce missed threats but may also increase unnecessary warnings. This was observed in PII/OpSec evaluation, where safe and ambiguous content produced weaker results compared with explicit or combined-risk cases.

False positives are not only a technical issue; they affect user trust. If users repeatedly receive warnings for harmless content, they may ignore future alerts. Therefore, future versions should include better confidence calibration, uncertainty-aware warnings, and user feedback loops.

F. Lack of Full Quantitative Evaluation for the Overall Guardian System

The project includes quantitative evaluation at the module level, but a single unified quantitative evaluation for the full Guardian system was not performed. This is a reasonable limitation because each module has different label spaces, datasets, objectives, and evaluation metrics. A single benchmark containing misinformation, hate speech, scams, phishing, PII, and OpSec risks together would require a custom multi-threat annotated corpus, which was outside the project scope.

Therefore, the final system-level evaluation is best interpreted as qualitative and scenario-based, while module-level evaluation provides the main quantitative evidence.

G. Privacy and Deployment Limitations

Although The Guardian is designed with privacy in mind, backend-based processing still requires user text to be transmitted to a server. The project requirements specify that user-submitted text should not be stored permanently and that communication should be encrypted using HTTPS. However, future deployment would require stronger privacy guarantees, such as on-device inference, encryption at rest, clear consent mechanisms, and transparent data handling policies.

H. Computational and Latency Constraints

Real-time browser interaction requires low-latency processing. This limits the size and complexity of models that can be used. While lightweight transformer models such as DistilBERT support faster inference, larger models or external fact-checking calls may introduce delay. Scam/phishing detection with URL expansion, blacklist checks, and safe preview rendering may also increase response time.

I. Multilingual and Local Context Limitations

The current system primarily focuses on English-language or English-dominant text. However, real social media users frequently use Sinhala, Tamil, code-mixed Sinhala-English, abbreviations, slang, and transliterated phrases. This is especially important for deployment in Sri Lanka. Future versions should incorporate multilingual datasets, Sinhala/Tamil NLP models, and localized threat patterns.

J. Ethical and Safety Limitations

The Guardian provides warnings and rewrite suggestions, but it must avoid becoming overly paternalistic or censorial. Since the system deals with sensitive content categories, it must also avoid unfair bias, demographic stereotyping, and excessive surveillance. For commercial deployment, transparency, opt-in consent, explainability, and user control would be essential.

Overall, The Guardian demonstrates strong potential as a practical human-layer cybersecurity system. However, the final prototype should be understood as a research system that validates modular AI-assisted risk detection, not a fully production-ready security product. Future improvements should focus on large-scale real-world evaluation, multilingual support, stronger privacy-preserving deployment, better calibration, and broader platform compatibility.

Table 10: Summary of Key Limitations and Future Mitigations

Limitation	Impact	Future Mitigation
Dataset bias and limited real-world data	Reduced generalization	Use larger, real-world, ethically sourced datasets
Synthetic OpSec dataset	Possible scenario coverage gaps	Expand with human-annotated real examples
No full quantitative system benchmark	Limits whole-system performance claims	Build a unified multi-threat benchmark
Edge cases such as sarcasm and coded language	False positives or missed detections	Add adversarial and edge-case datasets
Social media platform scope	Limited cross-platform applicability	Extend to WhatsApp, Telegram, email, Reddit, Discord
English-focused text	Weak multilingual coverage	Add Sinhala, Tamil, and code-mixed models
Backend processing	Privacy and trust concerns	Explore on-device or privacy-preserving inference
Warning fatigue	Reduced user trust	Add calibrated confidence and adaptive alerting

Model bias	Unequal performance across groups	Conduct fairness testing and bias mitigation
Limited user testing	Unverified real-world usability	Conduct user studies and surveys

10. Commercialization and Entrepreneurship Potential

This section presents the commercialization and entrepreneurship potential of The Guardian. The purpose of this section is not to claim that the prototype is already a market-ready product, but to demonstrate that the research has practical application potential beyond an academic setting.

A. Market Opportunity

The Guardian operates within the broader cybersecurity and human risk management market. The growth of social media usage has increased the attack surface for misinformation, hate speech, scams, phishing, and privacy leakage. DataReportal reported 5.24 billion active social media user identities worldwide in 2025, illustrating the scale of the potential user base [26]. At the same time, Gartner projected worldwide end-user spending on information security to reach USD 213 billion in 2025, with further growth expected in 2026 [27].

Existing tools such as antivirus software, safe browsing filters, platform moderation systems, and fact-checking services provide partial protection. However, many are reactive, platform-dependent, focused on single threat categories, or designed primarily for enterprise use. The original Guardian commercialization plan identified a gap in real-time user-side protection for individual users, especially where existing solutions focus on enterprises rather than everyday social media users.

B. Target Users

The potential target users for The Guardian include:

- Everyday social media users who need simple, real-time safety guidance
- Students and young adults who actively share content online
- Journalists, activists, and public figures exposed to doxing, harassment, and manipulation
- Professionals who may unintentionally disclose sensitive work or personal information
- Universities and schools seeking digital literacy and cyber-awareness tools
- Organizations looking for continuous human-risk awareness training

The system is especially relevant for users who are not cybersecurity experts but regularly interact with risky content online.

C. Value Proposition

The key value proposition of The Guardian is that it moves beyond simple detection and provides real-time, explainable, user-centered guidance. The commercialization plan describes Guardian as more than a PII detector, emphasizing real-time privacy risk detection, context-aware OpSec analysis, attacker-aware risk inference, and safe rewrite suggestions.

The main value points are:

- Real-time browser-based protection
- Multi-threat detection in one system
- Explainable warnings rather than vague alerts
- Non-intrusive guidance instead of censorship
- Safe rewrite suggestions to reduce risk while preserving user intent
- Modular architecture that allows new threat modules to be added later

This makes The Guardian suitable as both a consumer tool and an institutional awareness platform.

D. Business Model Potential

A freemium model is the most suitable initial commercialization pathway. This allows users to access basic protection for free while advanced features are offered through paid tiers.

Table 11: Proposed Commercialization Model

Tier	Target User	Features	Pricing Strategy
Free	General users	Basic PII/scam warnings, limited explanations	Free

Pro	Individual users	Full module access, OpSec analysis, safe rewrite, advanced explanations	Low-cost monthly subscription
Education	Schools/universities	Student cyber-awareness dashboard, reporting, guided learning	Institutional licensing
Enterprise	Organizations	Human-risk dashboard, API access, employee awareness analytics	Per-user or annual licensing
API Licensing	Cybersecurity vendors	Access to selected detection modules	Partnership/API pricing

This business model should be treated as a potential pathway, not a confirmed implementation plan.

E. Go-To-Market Strategy

A realistic go-to-market strategy can be developed in phases:

Phase 1: Research Prototype

The system remains a browser-extension prototype used for academic testing, demonstrations, and usability feedback.

Phase 2: Student and Privacy Community Adoption

The extension can be shared with students, cybersecurity learners, privacy-conscious users, journalists, and researchers for feedback.

Phase 3: Educational Deployment

The system can be adapted as a digital literacy tool for universities and schools. This is practical because the system teaches users why content is risky rather than simply blocking it.

Phase 4: Enterprise and API Expansion

Once validated, modules can be packaged as APIs or integrated into cybersecurity awareness platforms.

F. Competitive Advantage

The Guardian’s competitive advantage lies in its combination of modular AI detection, browser-based real-time interaction, contextual risk inference, and user-centered explanations. Many existing tools focus on a single category, such as malicious URLs or misinformation credibility, whereas The Guardian combines multiple human-layer threat categories into one framework.

Table 12: Competitive Positioning

Feature	Traditional Tools	The Guardian
Detects multiple human-layer threats	Limited	Yes
Browser-based real-time guidance	Limited	Yes
User guidance instead of censorship	Limited	Yes
PII + OpSec contextual reasoning	Rare	Yes
Scam/phishing URL + language analysis	Partial	Yes
Explainable warnings	Limited	Yes
Safe rewrite suggestions	Rare	Yes
Modular architecture	Limited	Yes

G. Feasibility and Cost Considerations

The project is feasible because it relies primarily on open-source tools, public datasets, synthetic datasets, and standard computing resources. Prior project planning identified that Python, FastAPI, Hugging Face Transformers, PyTorch, TensorFlow, and other planned tools are available under open-source licenses, while development can be carried out using personal laptops and university computing resources.

This makes the project suitable for low-cost prototyping. However, commercialization would require additional costs, including cloud hosting, model serving infrastructure, security testing, extension store compliance, legal review, customer support, and ongoing model maintenance.

H. Ethical and Regulatory Considerations

Commercialization must be approached carefully because The Guardian processes potentially sensitive user text. A commercial version would require:

- Clear opt-in consent
- Transparent privacy policy
- No unnecessary storage of user content
- Secure transmission and processing
- Bias and fairness testing
- User control over alerts and data handling
- Compliance with applicable privacy regulations

The system should remain advisory rather than coercive. The user must retain autonomy over whether to edit, ignore, or act on the warning.

I. Entrepreneurial Potential

The Guardian has strong entrepreneurship potential as a human-layer cybersecurity tool. It is positioned at the intersection of cybersecurity, AI, digital literacy, and usable security. Its modular design enables multiple commercialization routes, including a browser extension, educational platform, enterprise training tool, or API-based security service.

However, the current project should be presented as a research prototype with commercialization potential, not as a completed startup product. Further work is required before market launch, including large-scale usability testing, platform compatibility testing, privacy review, security hardening, and business validation.

Overall, The Guardian demonstrates clear potential for practical deployment. Its strongest commercial value lies in its ability to provide real-time, understandable, and non-intrusive protection for human-layer threats that are often overlooked by traditional cybersecurity tools.

11. Conclusion and Future Work

This research presented *The Guardian*, a modular AI-powered system designed to provide real-time protection against harmful and misleading content on social media platforms. The system integrates multiple detection modules—misinformation detection, hate speech detection, scam and phishing detection, and personal information and operational security (PII/OpSec) risk detection—within a unified and scalable framework.

The proposed system addresses a critical gap in existing solutions by focusing on human-layer cybersecurity, where threats arise from user interaction with content rather than traditional system vulnerabilities. By combining transformer-based models with rule-based and heuristic techniques, *The Guardian* enables the detection of both explicit threats and context-dependent risks in real time.

A key aspect of this work is the adoption of a unified, hybrid detection strategy that integrates statistical learning with contextual and rule-based reasoning. The evaluation results demonstrate that while transformer-based models provide strong baseline performance, they exhibit limitations in handling ambiguous and context-dependent scenarios. The integration of hybrid techniques and multi-module reasoning enhances system robustness and enables more effective detection of complex, real-world threats.

Another important contribution is the system's user-centric and non-intrusive design. Rather than restricting user actions, *The Guardian* provides real-time explanations, warnings, and actionable guidance, allowing users to make informed decisions. This approach promotes user awareness and aligns with emerging trends in explainable and human-centered artificial intelligence.

The modular architecture further supports scalability and flexibility, enabling the integration of additional detection modules and future enhancements without requiring significant structural changes. Additionally, the use of lightweight transformer models ensures that the system remains efficient and suitable for real-time deployment within browser environments.

Despite its effectiveness, several limitations remain. The performance of detection modules depends on the quality and diversity of training datasets, which may introduce bias and limit generalization across different contexts. Furthermore, detecting highly nuanced or context-dependent threats continues to be a challenging problem. Real-time processing constraints may also restrict the complexity of models that can be deployed.

Future work will focus on several directions:

- Enhancing model robustness using larger and more diverse datasets
- Incorporating multimodal analysis (e.g., images and videos)
- Improving explainability through advanced interpretation techniques
- Integrating adaptive learning mechanisms to personalize user feedback
- Expanding deployment to mobile and cross-platform environments

Overall, The Guardian demonstrates the feasibility of integrating advanced natural language processing techniques with user-centric design principles to address modern social media cybersecurity challenges. The system highlights the importance of combining multiple detection strategies and contextual reasoning within a unified architecture, contributing toward the development of practical and scalable human-layer cybersecurity solutions.

Reference List

- [1] D. M. J. Lazer *et al.*, “The science of fake news,” *Science*, vol. 359, no. 6380, pp. 1094–1096, Mar. 2018, doi: 10.1126/science.aao2998.
- [2] H. Allcott and M. Gentzkow, “Social media and fake news in the 2016 election,” *Journal of Economic Perspectives*, vol. 31, no. 2, pp. 211–236, 2017, doi: 10.1257/jep.31.2.211.
- [3] S. Vosoughi, D. Roy, and S. Aral, “The spread of true and false news online,” *Science*, vol. 359, no. 6380, pp. 1146–1151, Mar. 2018, doi: 10.1126/science.aap9559.
- [4] A. Schmidt and M. Wiegand, “A survey on hate speech detection using natural language processing,” in *Proc. 5th Int. Workshop Natural Language Processing for Social Media*, 2017, pp. 1–10.
- [5] D. Sahoo, C. Liu, and S. C. H. Hoi, “Malicious URL detection using machine learning: A survey,” *arXiv preprint arXiv:1701.07179*, 2017.
- [6] I. H. Sarker *et al.*, “Cybersecurity data science: An overview from machine learning perspective,” *Journal of Big Data*, vol. 7, no. 1, pp. 1–29, 2020.
- [7] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [8] V. Sanh *et al.*, “DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
- [9] K. Shu *et al.*, “Fake news detection on social media: A data mining perspective,” *ACM SIGKDD Explorations Newsletter*, vol. 19, no. 1, pp. 22–36, 2017, doi: 10.1145/3137597.3137600.
- [10] C. Boididou *et al.*, “Detection and visualization of misleading content on Twitter,” *International Journal of Multimedia Information Retrieval*, vol. 7, no. 1, pp. 71–86, 2018, doi: 10.1007/s13735-017-0143-x.
- [11] G. Pennycook and D. G. Rand, “The psychology of fake news,” *Trends in Cognitive Sciences*, vol. 25, no. 5, pp. 388–402, 2021, doi: 10.1016/j.tics.2021.02.007.
- [12] P. Badjatiya *et al.*, “Deep learning for hate speech detection in tweets,” in *Proc. 26th Int. Conf. World Wide Web Companion*, 2017, pp. 759–760.

- [13] M. Mozafari, R. Farahbakhsh, and N. Crespi, “A BERT-based transfer learning approach for hate speech detection in online social media,” in *Complex Networks and Their Applications VIII*, Springer, 2020, pp. 928–940.
- [14] O. K. Sahingoz *et al.*, “Machine learning based phishing detection from URLs,” *Expert Systems with Applications*, vol. 117, pp. 345–357, 2019, doi: 10.1016/j.eswa.2018.09.029.
- [15] N. Abdelhamid, A. Ayeshe, and F. Thabtah, “Phishing detection based associative classification data mining,” *Expert Systems with Applications*, vol. 41, no. 13, pp. 5948–5959, 2014, doi: 10.1016/j.eswa.2014.03.019.
- [16] S. C. Jeeva and E. B. Rajsingh, “Intelligent phishing URL detection using association rule mining,” *Human-centric Computing and Information Sciences*, vol. 6, no. 1, 2016.
- [17] A. K. Jain and B. B. Gupta, “Towards detection of phishing websites on client-side using machine learning based approach,” *Telecommunication Systems*, vol. 68, no. 4, pp. 687–700, 2018.
- [18] Information Commissioner’s Office, “What is personal data?” ICO, 2024. [Online]. Available: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/personal-information-what-is-it/what-is-personal-data/>
- [19] D. Gunning, “Explainable Artificial Intelligence (XAI),” Defense Advanced Research Projects Agency (DARPA), 2017.
- [20] S. Lewandowsky *et al.*, “Misinformation and its correction: Continued influence and successful debiasing,” *Psychological Science in the Public Interest*, vol. 13, no. 3, pp. 106–131, 2012, doi: 10.1177/1529100612451018.
- [21] S. Selvam and A. Ghosh, “PANORAMA: A synthetic dataset for privacy risk evaluation in online environments,” 2025. *(Use exact title from your PDF — update if needed)*
- [22] IBM Research, “Contextual privacy: Leveraging context for privacy risk understanding,” 2025. *(replace with exact title from your paper)*
- [23] TRACE-RPS Authors, “TRACE-RPS: Threat-aware contextual privacy risk scoring,” 2026. *(update title exactly from your file)*
- [24] PII-Bench Authors, “PII-Bench: Evaluating query-aware privacy protection systems,” 2025. *(update if needed)*

[25] M. Sokolova and G. Lapalme, “A systematic analysis of performance measures for classification tasks,” *Information Processing & Management*, vol. 45, no. 4, pp. 427–437, 2009, doi: 10.1016/j.ipm.2009.03.002.

[26] DataReportal, “Digital 2025: Global Overview Report,” 2025. [Online]. Available: <https://datareportal.com/reports/digital-2025-global-overview-report>

[27] Gartner, “Gartner forecasts worldwide end-user spending on information security to total \$213 billion in 2025,” Jul. 2025. [Online]. Available: <https://www.gartner.com/en/newsroom/press-releases/2025-07-29-gartner-forecasts-worldwide-end-user-spending-on-information-security-to-total-213-billion-us-dollars-in-2025>